

Yashas Bhat

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EDUCATION

University of Maryland

Bachelor of Science in Computer Science, Minor in Mathematics

GPA: 3.65/4.0

College Park, MD

08/2023 – 05/2027

SKILLS

Languages: Python, Rust, TypeScript, JavaScript, Java, SQL (Postgres), C/C++, OCaml

Frameworks & Tools: FastAPI, Flask, Axum, Next.js, React, Node.js, WebSockets, Tokio, FastMCP, Typer, LlamaIndex, Chroma, Git, Docker, Linux/UNIX, AWS

WORK EXPERIENCE

Amazon Web Services

06/2026 – 08/2026

SDE Intern

Cupertino, CA

- Enabled software CI for Nitro SSD recovery in the SSD emulator (MCE), cutting a 5–6 hour hardware-rig pass to ~8 minutes (~40×).
- Built deterministic fault injection (4 failure modes), a CLI, and 22 tests; lifted live-reset coverage from 24% to 91.5% and folded hardware-only failure paths into every code review.
- Reimplemented the C/C++ live-reset state machine so a hung card resets in place; host I/O continues through the reset (CRC32C-verified), adding +633 lines of existing FTL recovery coverage.

Children's National Hospital

08/2024 – 05/2025

Student Developer

College Park, MD

- Architected [TrachSense](#), a real-time airway-alert platform for tracheostomy patients, and led the TrachHub backend from sensor ingestion through a WebSocket streaming layer to mobile (**under 1 second** latency).
- Built a breathing-pattern analysis engine (hysteresis, RMS, standard deviation, flatline ratio) that cut false alerts by 10%.

Bharat Petroleum Corporation Limited

12/2024 – 01/2025

Software Engineering Intern

Mumbai, India

- Developed an anomaly detection framework combining DBSCAN clustering and Isolation Forest, **identifying 950+** potentially fraudulent distributors from **320K+** digital customer verification transactions and improving fraud detection accuracy over rule-based checks.
- Designed a feature extraction pipeline with 6 behavioral features, applying PCA visualization to surface irregular distributor behavior and weekday fraud trends for auditors.

Amazon Project Kuiper

01/2024 – 05/2024

Student Developer

College Park, MD

- Engineered 3 core APIs and the FastAPI/PostgreSQL backend for a blockchain-integrated [Space Safety Visualizer](#) on [Amazon Project Kuiper](#), enabling real-time collision prediction for 10+ satellite operators on a 13-engineer team.

PROJECTS

[Canvas MCP](#) | Python, FastMCP, Canvas API, Typer

07/2025

- Built an MCP server and CLI with 25+ Canvas tools so AI clients can pull live course data (assignments, grades, files) without custom API glue.
- Added URL resolution and course-context snapshots with Chrome-cookie, session-cookie, and API-token auth across MCP transports.

[Codexport](#) | Rust, Axum, Rusqlite, Comrak, Tokio

03/2026

- Developed a Rust CLI and local web app for browsing Codex chat history, rendering sessions as transcripts or clean Markdown, and exporting conversations for reuse and documentation.
- Added print-friendly PDF views and expiring share links through a Tailscale Funnel share server, turning chat history into portable artifacts.

[Know](#) | Python, LlamaIndex, Chroma

01/2026

- Created a local search CLI that watches directories, chunks 20+ file types, and stores a persistent Chroma index for semantic retrieval.
- Added incremental indexing, deduplication, and BM25/hybrid retrieval, cutting re-index time by 67% from 5.7s to 1.9s.